

Isaac Nathan da Silva Barbosa

COMPUTER ENGINEERING + AI — FULL-STACK · PYTHON · REACT · CLOUD

PWD — ASD (ICD-11: 6A02.2) + ADHD (ICD-11: 6A05.2)

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SUMMARY

Python · FastAPI · React 19 · Azure Cosmos DB · GPT-4.1 · Flutter · PyTorch — 10 months of professional experience. From Industrial Chemistry to Computer Engineering: built ForestAI from scratch (Stack Overflow + Thonny, no AI-assisted coding). At Paware, migrated hundreds of GB to Azure Cosmos DB (Meritage Homes, USA) and architected agentic AI pipelines for HelloSocial (GPT-4.1 tool calling, DALL-E 3, Flux). I learn by solving real problems — from cross-platform MIME types to ReAct agents with schema enforcement.

PROFESSIONAL EXPERIENCE

Full-Stack Developer with AI focus

Oct 2025 — May 2026

Paware Softwares

- Migrated hundreds of GB of legacy datasets from Google Drive to Azure Cosmos DB for Meritage Homes (USA) — pipeline with automated cookie-based extraction, compression, renaming, and injection into panels for WhatsApp agent embedding with strict file size limits. Containerized with Docker for reproducible environments and wrote schema validation layer with automated rollback.
- Resolved cross-platform MIME type issue (Android native vs iPhone requiring octet-stream — with extensive testing on iOS devices). This validation layer became the backbone of the final Azure Cosmos DB migration.
- Architected agentic image-generation pipeline for HelloSocial — GPT-4.1 agent with tool calling, up to 10 reasoning iterations, integrating Flux Kontext Pro + DALL-E 3 + Placid/Canva APIs. Fallback to SQLite/local PIL when paid APIs weren't needed.

Robotics Instructor

Sep 2025 — Oct 2025

SuperNerds

- Learned Arduino and LEGO in under 2 weeks and taught robotics to children/teens, connecting theory to real applications.
- Balanced mornings here with evening work at Paware.

ENGINEERING PROJECTS

HarpIA github.com/xAngryBadger/harpia

- Creative automation engine with 7+ AI models — GPT-4.1 agent with tool calling, autonomous agentic pipeline (copywriting, image search, design compositing, video generation). Lightweight stack by default (SQLite + local PIL) with fallback to paid APIs.
- 6,900+ LOC async Python with security tests and zero hardcoded secrets. Backend swap: local SQLite to Azure Cosmos DB + Blob Storage, toggled via env var. Cron-ready with file-locking and stuck batch recovery.

Python · GPT-4.1 · DALL-E 3 · Flux 2.0 Pro · Sora · Veo 3.1 · Azure Cosmos DB · SQLite

Flora Sensus github.com/xAngryBadger/flora-sensus

- Flutter offline-first app for forest inventory with custom sync engine — cascading remap detection through FK chain (Propriedade → UT → Parcela → Planta → Foto) and atomic rollback via Drift transactions.
- ~24K LOC — architecture and sync logic built from scratch; code generated with LLM assistance (web) and manually reviewed. React admin panel with auth, photos, reports, and XLSX/PDF/CSV export.
- Exponential backoff with jitter, auth retry wrapper with transparent 401 refresh, ngrok bypass for development.

Flutter · Dart · Drift/SQLite · React · PocketBase · TypeScript

ForestAI github.com/xAngryBadger/forestai

- Forest species detection and classification with Deep Learning — built from scratch without AI-assisted coding. Stack Overflow + Thonny IDE only.
- Manual annotation of hundreds of drone images from Fundação Renova (bounding boxes), DeepForest/YOLO training on local GPU, stratified splits. TensorBoard curve interpretation — detecting memorization vs generalization. The hard way built the intuition that made every subsequent framework click faster.

PyTorch · DeepForest · OpenCV · scikit-learn · TensorBoard

Fennec Excel github.com/xAngryBadger/Sahara-Fenneck

- Local AI assistant for Excel via ReAct agent (Ollama/qwen2.5). Natural language commands to filter, sort, rename sheets and manipulate spreadsheets with auto-checkpoint before every change.
- 6+ OAuth integrations (Gmail, Teams, Calendar, Drive, Outlook, Trello) with user confirmation before modifications.
- Hand-crafted visual design with pastel palette and original mascot (Fennec). Native Windows installer (Inno Setup + PyInstaller). Bilingual PT/EN interface.

Python · Ollama · CustomTkinter · xlwings/COM · PyInstaller · Inno Setup

Orca github.com/xAngryBadger/orca

- Operational planning engine for large-scale forest restoration — automatic generation of executive dossiers with crew allocation, territory mapping, and schedules.
- Tariff and operational cost management. NiceGUI + Rich CLI interface with unit test suite.

Python · pandas · NiceGUI · Rich CLI · openpyxl · unittest

MaineCoon github.com/xAngryBadger/minepal

- Minecraft bot with natural language commands via LLM (NVIDIA NIM API) — mine, craft, follow, navigate, and interact via chat. Reinforcement learning module for autonomous behavior.

Node.js · mineflayer · NVIDIA NIM API · Reinforcement Learning

HelloSocial

- AI-powered social media post creation and scheduling platform — project at Paware that inspired HarpIA. Image generation pipeline with Flux Kontext Pro and DALL-E 3, copy and template agents. PIX via Mercado Pago (AguaQuality). Canva + Placid (HelloSocial/Paware). Pexels API for automatic image search in HarpIA.

Python · FastAPI · Azure OpenAI · Flux · Canva API · React

Capivara github.com/xAngryBadger/capivara

- Full PDF suite with 15 tools — DOCX/XLSX → PDF, PDF → DOCX, compress, merge, split, rotate, watermark, page numbers, header/footer, protect, unlock, OCR, PDF → images, PDF/A. React 19 frontend with hash routing and tool grid, FastAPI backend with StreamingResponse.

- Free serverless tunneling via cloudflared + Google Colab — zero account, zero token. Lazy imports for heavy deps (PyMuPDF, pytesseract, pikepdf). pypdf 6+ compat (PdfWriter-based merge).

React 19 · TypeScript · FastAPI · pypdf · reportlab · PyMuPDF · cloudflared

Tarsier github.com/xAngryBadger/tarsier

- 100% client-side JSON editor with CodeMirror 6, virtualized tree, ReactFlow graph, diff mode, TypeScript generator and integrated REST client. Zero backend — everything runs in the browser.

React 19 · CodeMirror 6 · ReactFlow · TypeScript · Vite

Kakapo github.com/xAngryBadger/kakapo

- Client-side image editor with Canvas API — compress, resize, crop, filters, rotation and adjustments. Commit-based editing engine with undo/redo. 100% in-browser, zero uploads.

React 19 · Canvas API · TypeScript · Vite

Oilbird github.com/xAngryBadger/oilbird

- Markdown → PDF converter with real-time preview and customizable themes. FastAPI + WeasyPrint backend for professional PDF rendering, cloudflared tunnel via Google Colab.

React 19 · TypeScript · FastAPI · WeasyPrint · cloudflared

Cegonha github.com/xAngryBadger/cegonha

- Resume generator with editable templates, real-time preview and PDF export. Dynamic forms with drag-reorder. Sage green palette with editorial typography.

React 19 · TypeScript · Vite · Tailwind CSS v4

EDUCATION

Industrial Chemistry

2022

UFOP

FIRST STEPS IN PYTHON — THONNY IDE

Chemical Engineering

2022 — 2024

UFSJ — Campus Alto Paraopeba

TRANSITION PERIOD — SELF-DIRECTED AI STUDY

Computer Engineering

2024 — 2029

Cruzeiro do Sul

5TH SEMESTER — IN PROGRESS

CORE STACK

FRONTEND

React 19 · TypeScript · JavaScript · Tailwind CSS v4 · Flutter · Vite · React Native · Expo

React 19 + TypeScript → HarpIA frontend, Flora Sensus admin panel, Inovesa institutional site, this very portfolio.

BACKEND

Python · FastAPI · Node.js · Express · Java · SQL

FastAPI + Azure Cosmos DB → HarpIA pipeline, AguaQuality IoT backend, HelloSocial (Paware). Python async as production language.

CLOUD & INFRA

Azure · Azure Cosmos DB · Docker · Git / GitHub · PocketBase

Database migration to Azure Cosmos DB (Meritage Homes, USA). Docker for AI agents. PocketBase as backend for Flora Sensus.

AI & DATA

PyTorch · DeepForest · Azure OpenAI · DALL-E 3 · Flux · GPT-4.1 (Agentes) · NVIDIA NIM API · Gemini SDK · scikit-learn · OpenCV · Ollama

7+ AI models integrated in HarpIA. ReAct agent with Ollama in Fennec Excel. Species detection with PyTorch + DeepForest in ForestAI.

INTEGRATIONS

Mercado Pago PIX · Canva Connect API · Placid API · Templated.io · Pexels API · Sora / Veo (Video AI)

PIX via Mercado Pago for AquaQuality. Canva Connect + Placid for HelloSocial (Paware). Pexels API for automatic image search in HarpIA.

OTHER

Cybersecurity · Agile Methodologies · Testes de API · Fluent English · NiceGUI · PyInstaller / Inno Setup · xlwings / COM · Linux (CachyOS/Hyprrland)

Fluent English for documentation and meetings with US team. Orca with NiceGUI + Rich CLI. PyInstaller + Inno Setup for Fennec installer. Custom CachyOS/Hyprrland rice.

CERTIFICATIONS

- **Python Essentials 1** — Cisco Networking Academy *Programming fundamentals with Python* — logic, structures, and first scripts.
- **Python Essentials 2** — Cisco Networking Academy *Advanced Python expansion* — OOP, libraries, and certification prep.
- **Data Science Essentials with Python** — Cisco Networking Academy *Data analysis with Pandas and Matplotlib* — hands-on, project-based learning.
- **Data Analytics Essentials** — Cisco Networking Academy *Essential analytics tools recognized by the industry*.
- **AI for Process Optimization & Decision-Making** — Escola Virtual Gov · Enap · Serpro (71h) *Program by the Gov AI Nucleus (PBIA)* — strategic use of AI in public management, data analysis, and information security.
- **Networking Basics** — Cisco Networking Academy (120h) *Completed during Computer Engineering at Cruzeiro do Sul*
- **Intro to Cybersecurity** — Cisco Networking Academy *Foundation in network security and cyber threats*
- **IT Security** — Bradesco Foundation *Complement in corporate infrastructure and data protection*
- **Fluent English** — KUMON (3 years) *Enables technical documentation and meetings with international teams*

LANGUAGES

- Portuguese — Native
- English — Fluent (KUMON, 3 years)